Integration Project – Model Predictive Control (MPC)
Modeling and Control of a Transport Network

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1 Introduction

In zone-based evacuation planning, the region to evacuate is divided into zones, and each zone must be assigned a path to safety and departure times along the path. Zone-based evacuations are highly desirable in practice because they allow emergency services to communicate evacuation orders and to control the evacuation more precisely. Zone-based evacuations may also be combined with contraflows (to maximize the network capacities) and may impose additional constraints on the evacuation path (e.g., path convergence) and the departure times (e.g., non-preemption). This project focuses on the modeling and optimal control of a transport network represented as a directed graph. The objective is to describe the evolution of the volumes stored in the different components of the network and to formulate a Model Predictive Control (MPC) problem in order to optimize the global behavior of the system over a finite time horizon.

First, a mathematical model of the network dynamics is introduced, together with the associated physical constraints. Then, a convex relaxation of the model is proposed in order to make the optimal control problem tractable. Finally, a finite-horizon optimal control problem is formulated.

2 Network Model

The transport network is represented by a directed graph $G = (V, E)$ composed of two types of nodes:

- $\mathcal{T}$: set of **tanks** ,
- $\mathcal{X}$: set of **transport cells** .

The set of vertices is given by $V = \mathcal{T} \cup \mathcal{X}$, with $\mathcal{T} \cap \mathcal{X} = \emptyset$. The evolution of the evacuation network is described in discrete time. Time is discretized with a constant sampling period $h > 0$:

$$t_k = kh, \quad k = 0, 1, \ldots, K.$$

At each instant, the state of the system is characterized by the number of vehicles stored in each tank x_T and in each cell x_i. Control decisions mainly act on the tanks through control variables u_T, which regulate the number of vehicles injected into the network. Interactions between the different elements are described by flows f_{ij}, representing the number of vehicles moving from cell i to cell j, while the final exits of the network are represented by the variables μ_i associated with terminal cells. This formulation allows for precise tracking of vehicle propagation throughout the network, accounts for congestion effects, and provides a basis for a model predictive control problem aimed at optimizing evacuation over a finite time horizon under physical and operational constraints.

2.1 State and control variables

For each discrete time step k, the following variables are defined:

- $x_T(k)$: number of cars stored in tank $T \in \mathcal{T}$,
- $x_i(k)$: number of cars stored in cell $i \in \mathcal{X}$,
- $u_T(k) \in [0, 1]$: control associated with tank T,
- $f_{ij}(k) \geq 0$: flow from node i to node j,
- $\mu_i(k)$: outflow leaving the network from terminal cells.

2.2 Demand and supply functions

The dynamics of vehicle flow within each transport cell are described using demand and supply functions. The **demand function** represents the potential outflow from a cell based on the number of vehicles currently present, the cell's length, and its velocity, while allowing modulation through a control : we assume that we can control the speed of the vehicles. In contrast, the **supply function** models the capacity of a cell to accept incoming vehicles, which decreases as the cell becomes congested. Together, these functions account for congestion effects and flow limitations, ensuring that vehicle movements remain physically realistic and that control actions respect the network's capacity constraints. This formulation is essential for designing an effective model predictive control strategy, as it provides a clear link between the state of the network (vehicle densities) and the possible flows between cells.

Demand function The demand function is defined as:

$$d_i(x_i(k)) = \gamma_i \min\left(\frac{v_i}{l_i}x_i(k),\ C_i\right),$$

where v_i denotes the velocity, l_i the length of the cell, C_i the maximum capacity, and $\gamma_i \in [0,1]$ is a control parameter used to regulate the flow.

Supply function The supply function is given by:

$$s_i(x_i(k)) = \max\left(c_i - w_i x_i(k),\ 0\right),$$

where c_i is a capacity constant and w_i is a saturation coefficient.

2.3 Flow definitions

The vehicle flows in the network are defined to capture physical limitations and capacity constraints. The **flow between two cells** is limited by the demand of the upstream cell and the supply of the downstream cell, ensuring realistic traffic propagation. When multiple cells converge into a single cell, the incoming flow is distributed proportionally to the demand of each upstream cell while respecting the capacity of the downstream cell. The **flow from a tank to a cell** is controlled by the tank's control variable and is limited by the supply of the target cell. For terminal cells with no successors, the **outflow from the network** corresponds directly to the demand of the cell. Finally, flows from transport cells back to tanks or upstream cells are prohibited, reflecting the irreversibility of the evacuation. These definitions link the network state and control decisions to the actual flows between cells and tanks, while respecting congestion and capacity constraints.

Cell-to-cell flow For two transport cells $i, j \in \mathcal{X}$, the flow is defined as:

$$f_{ij}(k) = \min\left(d_i(x_i(k)),\ s_j(x_j(k))\right).$$

Many cells to a single cell flow We have $m \in \mathcal{X}$ cells that all lead to a single cell $j \in \mathcal{X}$. The flow is defined as :

$$\forall i \in [1, m],\ \ f_{ij} = \min\ \left(\beta d_i(x_i(k)), s_j(x_j(k))\right)$$

$$\text{with}\, \beta = \min(\frac{s_j(x_j(k))}{\sum_{i=1}^{m} d_i(x_i(k))}, 1)$$

Tank-to-cell flow For a tank $T \in \mathcal{T}$ and a cell $j \in \mathcal{X}$:

$$f_{Tj}(k) = \min\left(u_T(k),\ s_j(x_j(k))\right).$$

$$\text{with}\, u_T(k)\ \leq\ x_T(k)$$

Outflow from the network If a cell i has no successor, the outflow leaving the network is:

$$\mu_i(k) = d_i(x_i(k)).$$

One-way evacuation Reverse flows from transport cells back to tanks or upstream cells are forbidden:

$$f_{jT}(k) = 0, \quad \forall j \in \mathcal{X},\ \forall T \in \mathcal{T}.$$

$$f_{ji}(k) = 0, \quad \forall i, j \in \mathcal{X},\ j \geq i.$$

2.4 Discrete-time system dynamics

The discrete-time dynamics of the network describe the evolution of the number of vehicles in tanks and transport cells. Tank number of vehicles decrease according to the outgoing flows to the cells, while cell number of vehicles increase with incoming flows and decrease with outgoing flows and network outflows. This formulation directly links the flows to the network states.

Tank dynamics The evolution of the tank number of vehicles is described by:

$$x_T(k+1) = x_T(k) - h \sum_{j:\, T \to j} f_{Tj}(k).$$

Cell dynamics The dynamics of the transport cells are given by:

$$x_i(k+1) = x_i(k) + h \left(\sum_{j:\, j \to i} f_{ji}(k) - \sum_{m:\, i \to m} f_{im}(k) - \mu_i(k) \right), \quad i \in \mathcal{X}.$$

2.5 Physical constraints

The system is subject to the following physical constraints. Vehicle volumes in tanks and cells must be non-negative, and flows between cells cannot be negative, as it is impossible to have a negative number of vehicles moving through the network. The control variable u_T, representing the number of vehicles leaving a tank, cannot exceed the volume available in that tank, ensuring that no more vehicles are evacuated than are present. Additionally, when multiple cells converge into a single cell j, the total incoming flow cannot exceed the supply of the destination cell, which enforces the network's physical limits.

The system is subject to the following constraints:

$$\begin{cases} x_i(k) \geq 0, \quad x_T(k) \geq 0, \\ f_{ij}(k) \geq 0, \\ 0 \leq u_T(k) \leq x_T(k). \end{cases}$$

When we get $m \in \mathcal{X}$ cells to a single cell $j \in \mathcal{X}$, we add these constraints :

$$\sum_{i=1}^{m} f_{ij} \leq s_j$$

3 Open-Loop Analysis of a Simple Network

3.1 First Network: Line Graph — Two Tanks and Six Cells

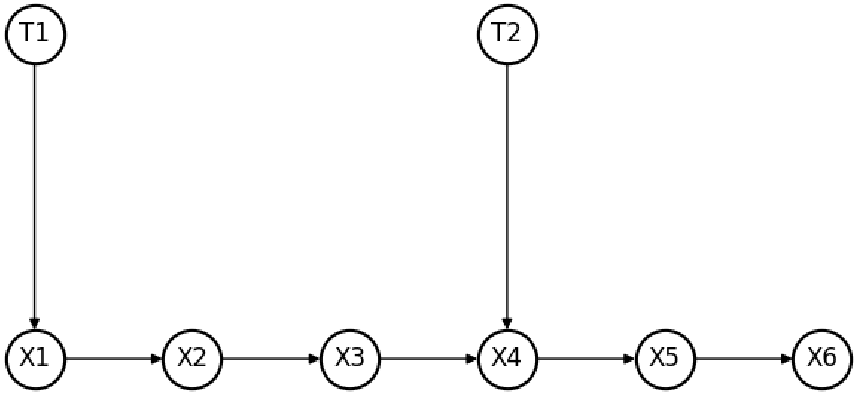

Figure 1: Network with two Tanks and six Cells

First we modeled a simple line graph network with 6 cells and two tanks plugged respectively on the first and the fourth cell as we can see in figure 1. Each cell has a 200 meters length and a capacity of 1000 vehicles/h. Each vehicle moves at 70 km/h in the cells. The two tanks are initially filled with 100 vehicles.

3.2 Second Network: Y-shaped Graph — Two Tanks and Six Cells

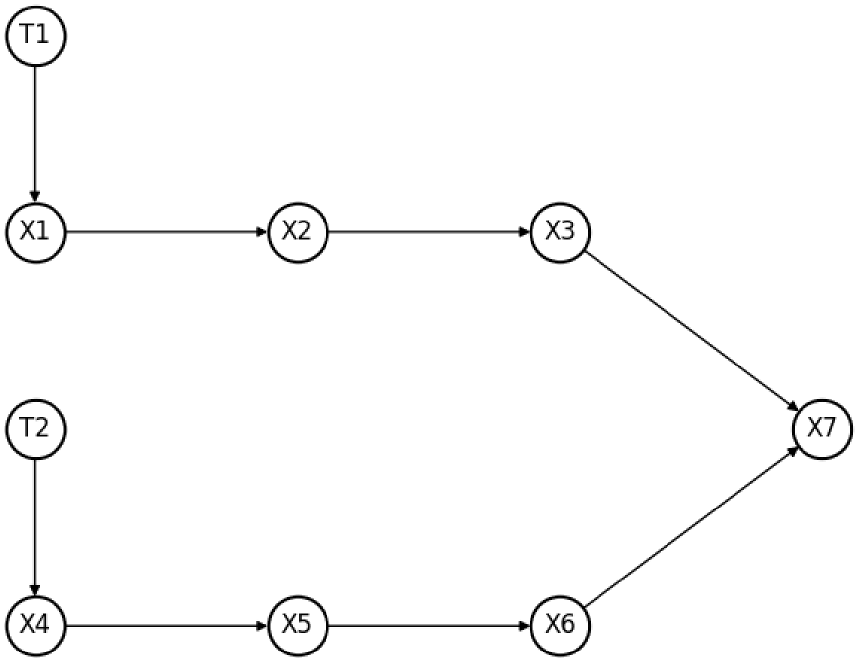

Figure 2: Network with two Tanks and six Cells

Then, we modeled a Y-shaped graph network with 7 cells and two tanks plugged respectively on the first and the fourth cell as we can see in figure 2. Each cell has a 200 meters length and a capacity of 1000 vehicles/h. Each vehicle moves at 70 km/h in the cells. The two tanks are initially filled with 100 vehicles.

3.3 Open-Loop Simulation Setup and Results

To empty the tanks, two naive open-loop strategies were tested. On one hand, both tanks were emptied simultaneously and, on the other hand, both tanks were fully emptied one after the other, beginning with the second tank. Both strategies were simulated for each of the networks and compared in figure 3 and figure 4. The control variable u_T was set to the constant value of 100 vehicles so that it did not constrain tanks'outflows. Therefore, these simulations correspond to the best evacuation time for both strategies, in the case of a constant control variable.

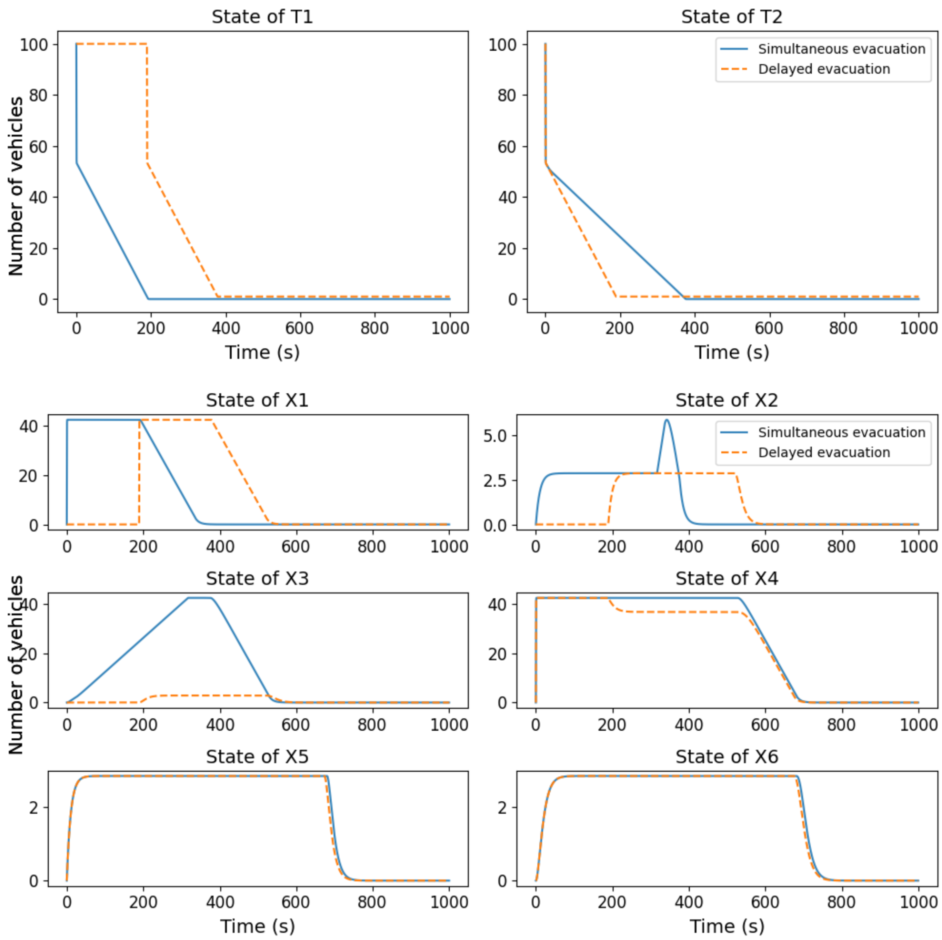

Figure 3: Two different evacuation strategies tested for the line graph : simultaneous or delayed evacuation

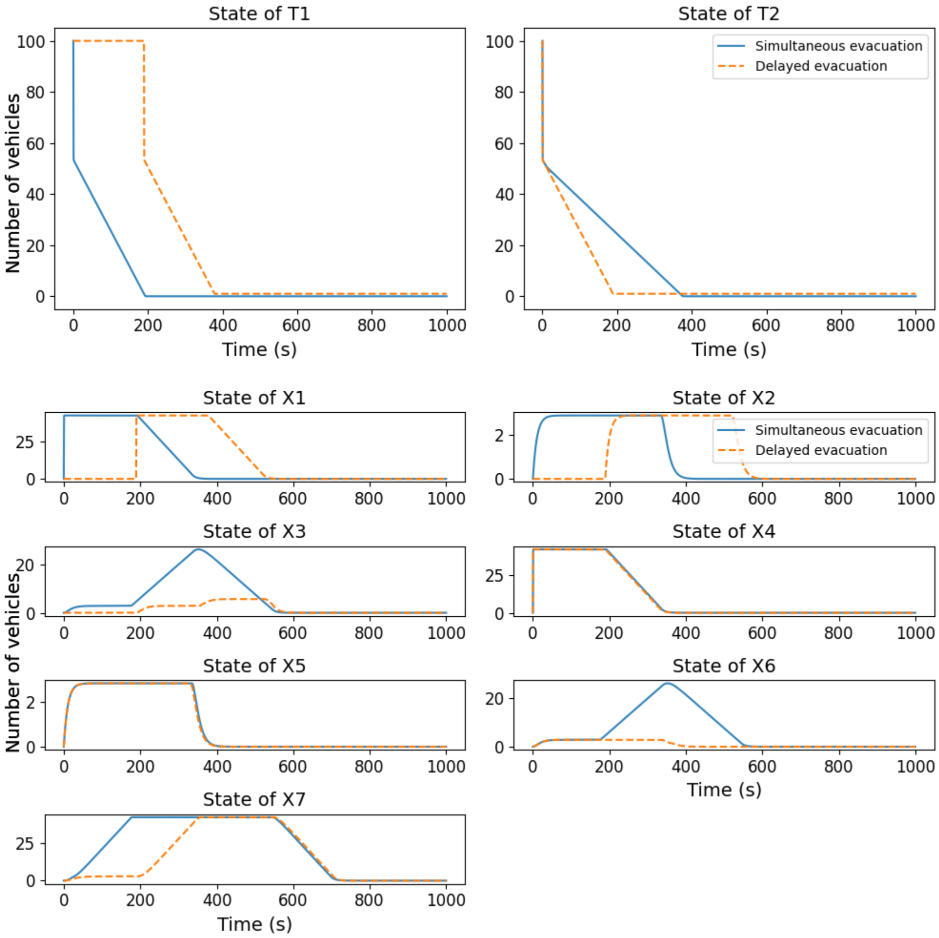

Figure 4: Two different evacuation strategies tested for the Y-shaped graph : simultaneous or delayed evacuation

When both tanks are emptied simultaneously in the line graph, there is more stress on cell X4 than for delayed evacuation. This can be explained by the addition of vehicles coming from the first and second tanks, which accumulate at cell 4. In the Y-shaped graph, the same results are observed. There is indeed more stress on cell X7 as this is where both branches merge but also on cells X3 and X6, the closest cells to X7. This is due to the "back-propagation" of saturation on transport cells, i.e. when a cell saturates, the straight upstream cells cannot evacuate their vehicles so they store more of them and end up saturating too. Besides, the saturation limit of each cell is about 42.5 vehicles, corresponding to the maximum number of vehicles that a 200m long road portion can contain (considering average vehicle length of 4.7m). Note that only one lane is used here for evacuation.

In the line graph, cells X5 and X6 do not store more than three vehicles. This is due to the absence of tanks or merge downstream so they can evacuate vehicles more easily as cell capacity is not a constraint anymore.

Furthermore, note the symmetry of the Y-shaped graph, causing it to show a symmetric behaviour on both branches in the case of simultaneous evacuation as the control variable is also identical.

Regarding the performance of each strategy, delayed evacuation is slightly quicker than simultaneous one in the line graph configuration, certainly because it reduces saturation. However, it is less suitable to the Y-shaped graph. This can be explained because delayed evacuation does not take advantage of the two parallel branches.

4 Model Predictive Control Formulation

The original flow model relies on $\min(\cdot)$ operators between demand and supply functions. It is these $\min(\cdot)$ operators that introduce nonlinearities and make the constraints non-convex. Such non-convexity prevents the use of efficient optimization methods within a model predictive control framework.

4.1 Convex Relaxation of the Flow Model

To ensure convexity, the $\min(\cdot)$ operators are replaced by inequality constraints, and the flows are treated as independent decision variables $\bar{f}_{ij}$. This relaxation yields a convex optimization problem while preserving the essential physical limitations of the network.

$$0 \leq \bar{f}_{ij}(k) \leq d_i(x_i(k)),$$

$$\sum_{j:\, j \to i} \bar{f}_{ji}(k) \leq s_i(x_i(k)).$$

4.2 MPC Problem Statement : Finite-horizon optimal control problem

A prediction horizon of length K is considered, together with the following cost function:

$$J = \sum_{k=0}^{K} \left(\sum_{i \in \mathcal{X}} x_i(k) + \sum_{T \in \mathcal{T}} x_T(k) \right).$$

The objective is to minimize J subject to:

- control constraints:
$$0 \leq u_T(k) \leq x_T(k), \quad \gamma_i \in [0, 1],$$
- flow constraints:
$$\sum_{m:\, i \to m} \bar{f}_{im}(k) \leq d_i(x_i(k)),$$
$$\sum_{j:\, j \to i} \bar{f}_{ji}(k) \leq s_i(x_i(k)),$$
$$\sum_{j:\, T \to j} f_{Tj}(k) \leq u_T(k)\, x_T(k),$$
- system dynamics,
- no reverse flows,
- given initial condition $x(0)$.

4.3 Numerical Implementation of the MPC Controller

The numerical solution to the MPC problem was performed using the cvxpy package, an open source Python-embedded modeling language for convex optimization problems, and the GUROBI solver to increase computational power.

5 Coupling the MPC Controller with the Simulator

5.1 Simulator Description

5.2 Closed-Loop Control Architecture

The control is done following these steps :

- Formulate the problem statement (cost function, constraints)
- Solve the optimization problem
- Apply the first half solutions of the optimization problem
- Go again with the previous steps until we get the last iteration

5.3 Closed-Loop Simulation Results

5.3.1 Junction with two single cells

This part is fed with both the MPC results and the simulator outputs based on the MPC solution. The MPC is run over a finite horizon of 200 for 2000 iterations.

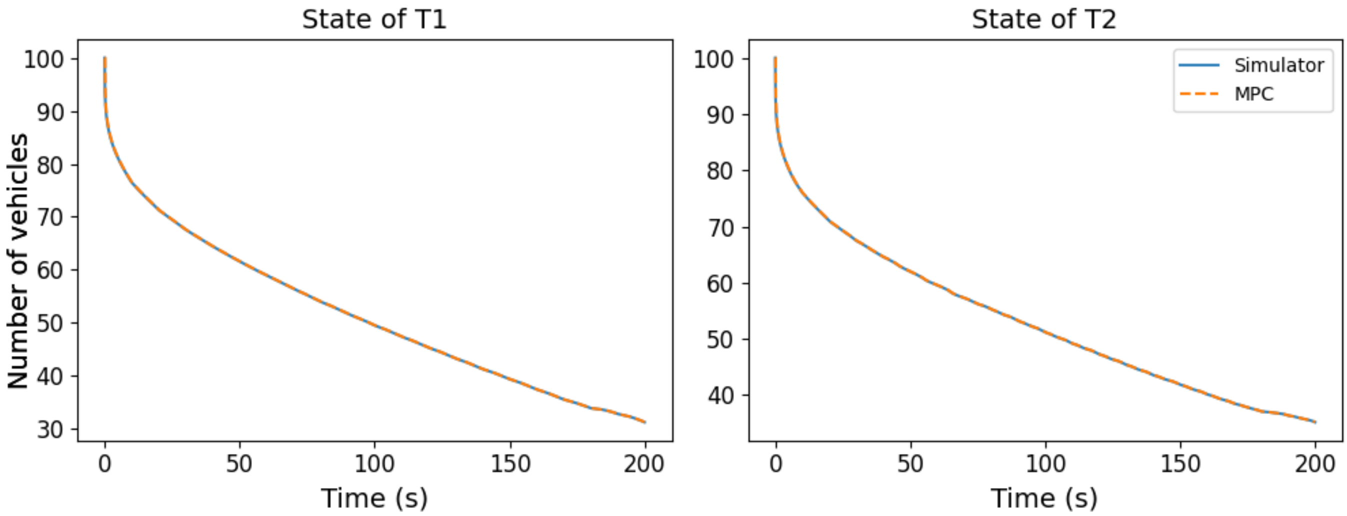

Figure 5: MPC tank states

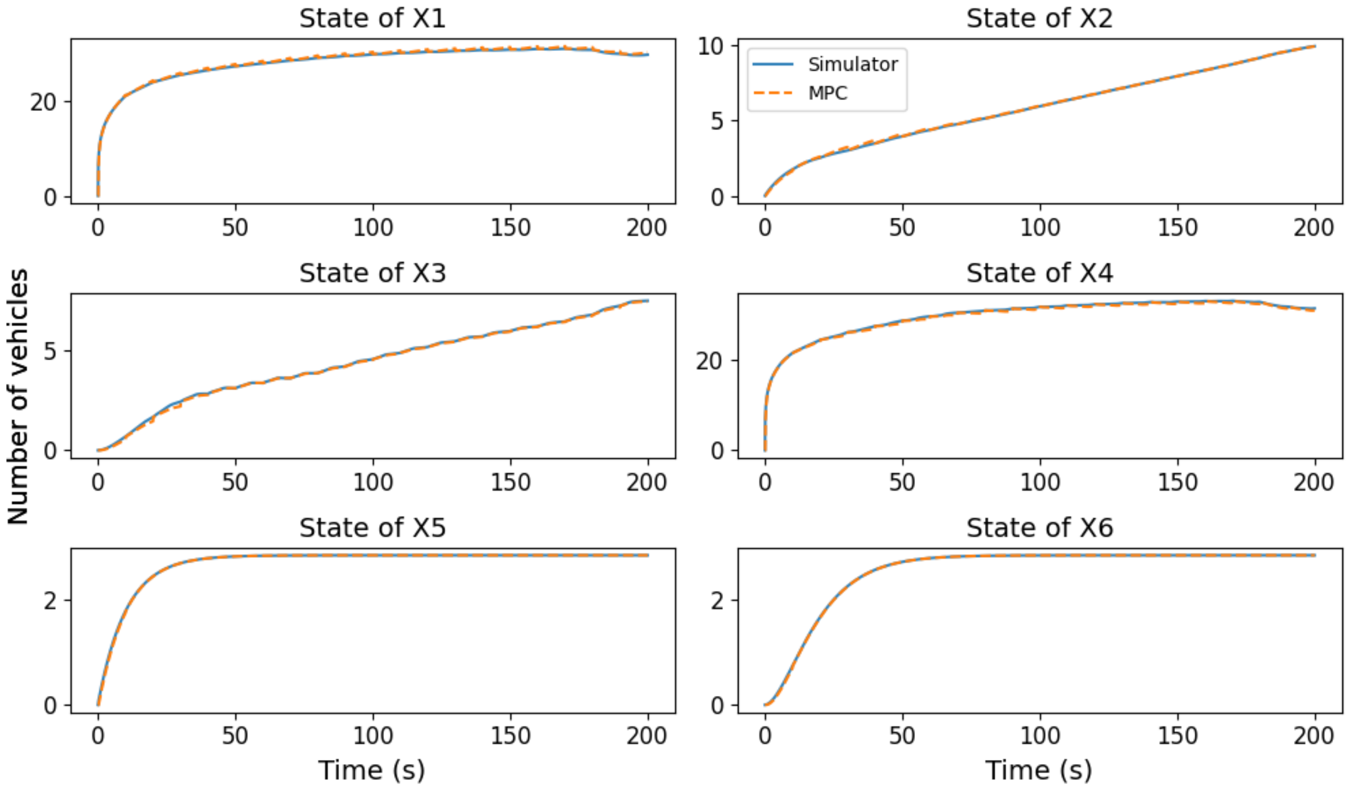

Figure 6: MPC cells states

We got in the simulator that we perfectly match the MPC plots using MPC solution. Especially the plots of the tanks and the cells x_1, x_4, x_5 and x_6. But we the others, it's approximately the same but we got good results indeed.

5.3.2 Junction with many cells to a single one

6 Results and Performance Analysis

6.1 Comparison Between Open-Loop and Closed-Loop Behavior

Table 1 compares the time performance of different control strategies. The results show that Model Predictive Control (MPC) with a prediction horizon of 800 and the optimization-based approach lead to very similar outcomes, with times of 733 s and 732 s respectively, indicating nearly identical closed-loop behavior. In contrast, open-loop strategies result in significantly longer times. When the two tanks empty sequentially, the time increases to 902 s, while it reaches 836 s when both tanks empty simultaneously. These results clearly highlight the advantage of closed-loop control strategies, which provide a more efficient dynamic management of the system and significantly reduce the time required to reach the condition where only one vehicle remains in zone X6.

Used Method	Time (in seconds) when there are only one vehicle left in X6
MPC with horizon 800	733
Optimization problem	732
Open-loop when the two tanks empty one after the other	902
Open-loop when both tanks empty at the same time	836

Table 1: Comparison Between Open-Loop and Closed-Loop Behavior

6.2 Impact of MPC on Congestion Reduction

6.3 Discussion on Model-Simulator Mismatch

7 Conclusion and Perspectives

- Summary of the main results
- Limitations of the proposed approach
- Perspectives and future work

References

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List of Figures